

TEAM PRXFR · IQOO CITY BATTLES 2026

WorldJam

Record the real sound of anything around you. On-device AI turns it into music.

Zero bundled sounds. No network. Playable with your eyes closed.

Android · React Native

Gemma 4 on-device

Native C++ audio



Music apps hand you someone else's sounds

The same sample packs

Every beginner opens the same library and makes something that sounds like everyone else's.

A studio you must learn

A DAW is a wall of controls. The idea dies before the first bar is finished.

AI that sends audio away

Generative tools upload your recordings and need a connection to answer.

Meanwhile every room already has instruments in it: a ceramic cup, a wooden table, a set of keys.

THE IDEA

Anything can be a studio

- 01 Capture** — Hold and strike an object. The phone records its real sound and measures how bright, how long and how pitched it is.
- 02 Direct** — Pick a vibe, or type what you want — an artist, a feeling, an instrument. Sing, and the melody becomes the song.
- 03 Compose** — Gemma, on the phone, arranges your sounds into a structured track with verses, builds and fills.

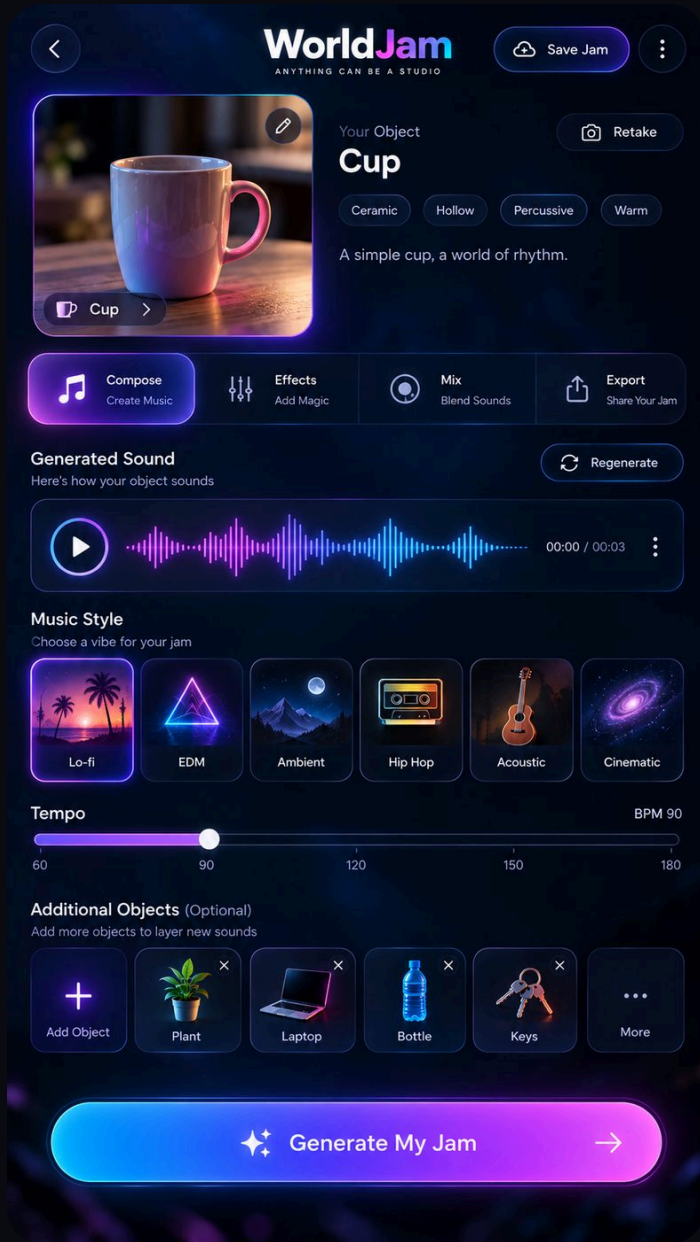


Four steps, one screen each



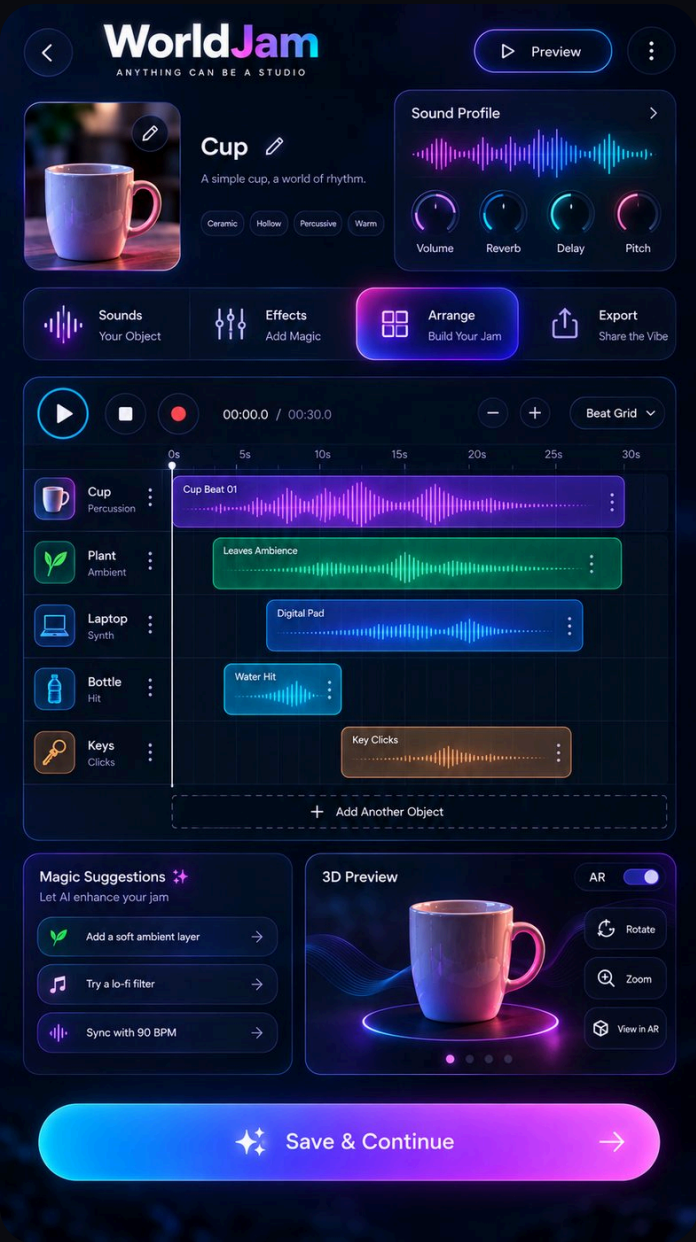
1 · Scan

Point, hold, strike. The real sound is captured.



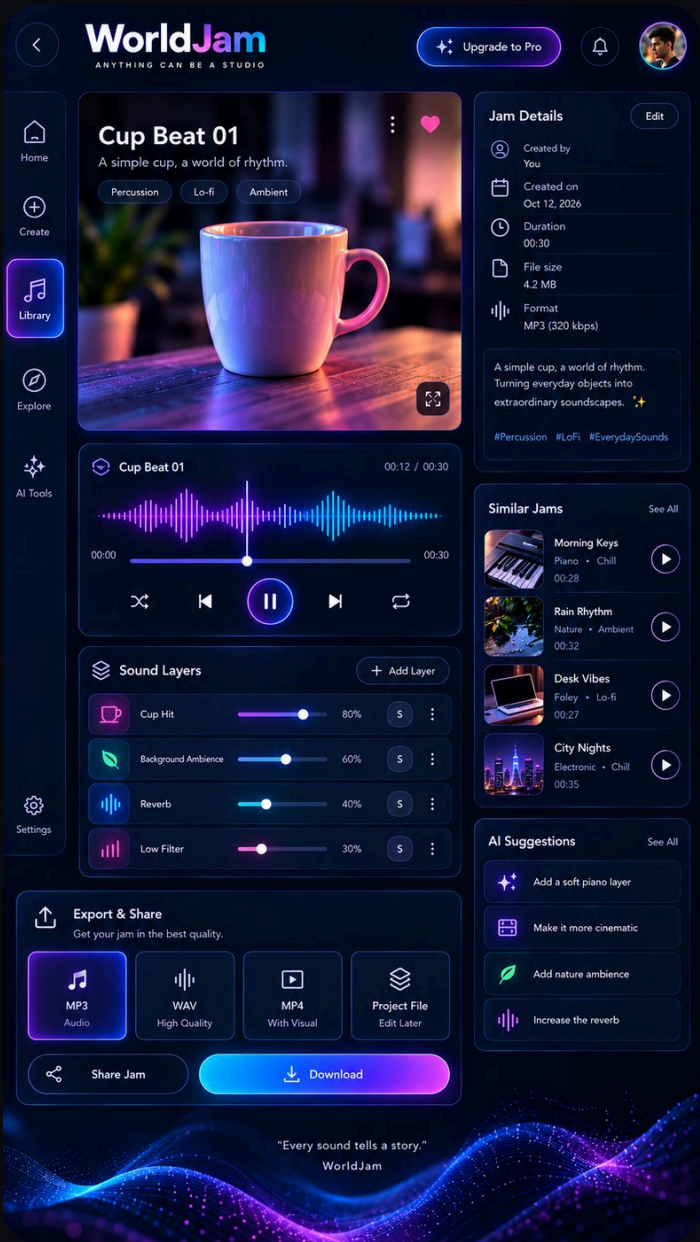
2 · Customize

Vibe, tempo, and a box that takes plain English.



3 · Compose

The arrangement, laid out on a timeline.



4 · Your Jam

Transport, layer mixer, export.

WHY THIS MATTERS MOST

Music has no boundaries. Neither should the instrument.

For a blind user, WorldJam is not a music app with accessibility bolted on. It is a way to **hear the room**: the phone names what is in front of you, confirms every capture aloud, and places each object in the stereo field where it actually sits. The mix becomes a map of your surroundings — and then it becomes a song.

Sense the surroundings

Spoken guidance describes what is nearby and confirms each recording, so a track can be built entirely by touch and sound.

Spatial by default

Objects are panned to their real position. Hearing the mix tells you where things are, not just how they sound.

Connect through it

A jam needs no sight, no reading and no shared language. It is the least gated way two people can make something together.

Every control carries a screen-reader label and role — including the knobs and faders, which adjust by gesture. An instrument you must watch is not an instrument.

NOT A MOCKUP

Running on the handset today

These are screenshots from the device, beside the design they were built to. Capture, playback, arrangement, sung-melody analysis, lyrics, export and session saving all work now.

389

tests
passing

24.8

ms tap to
sound

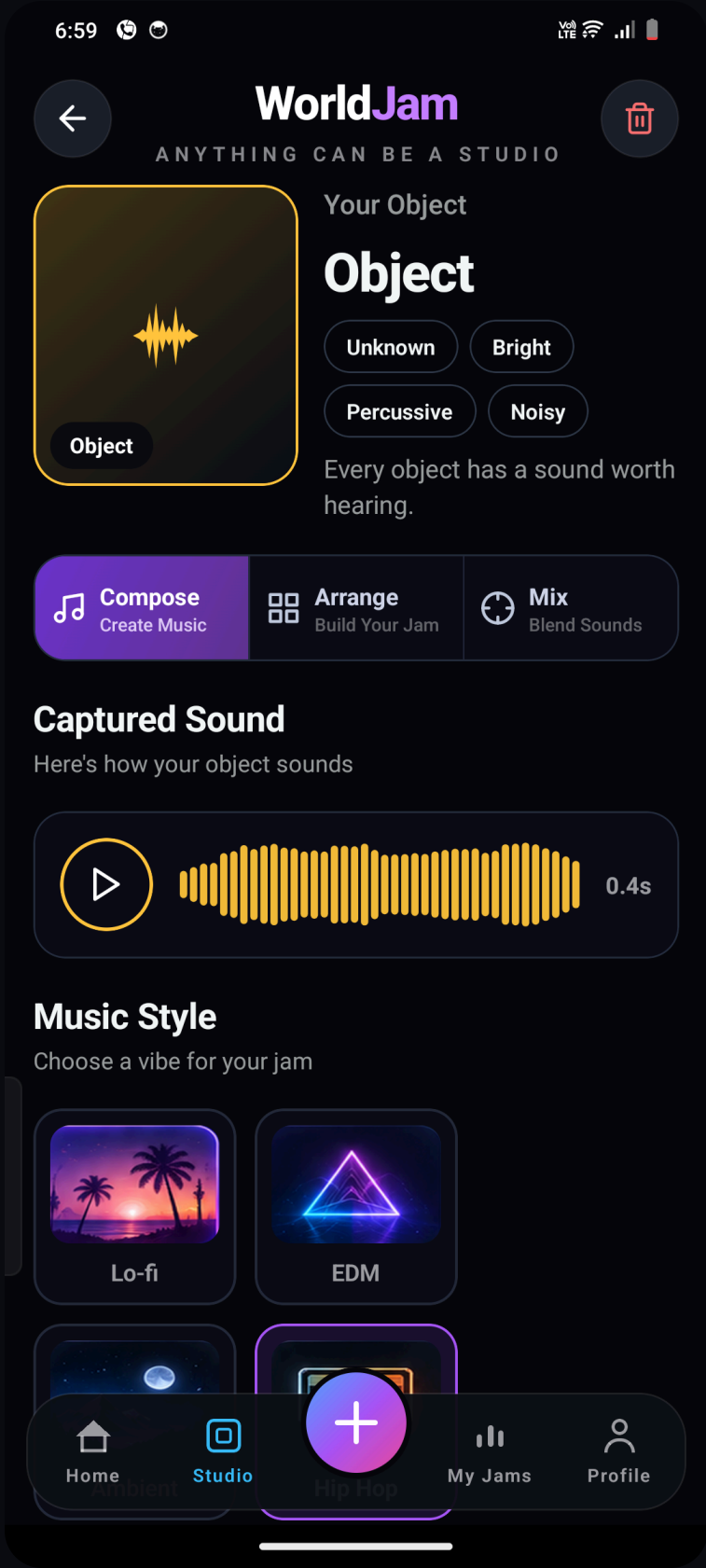
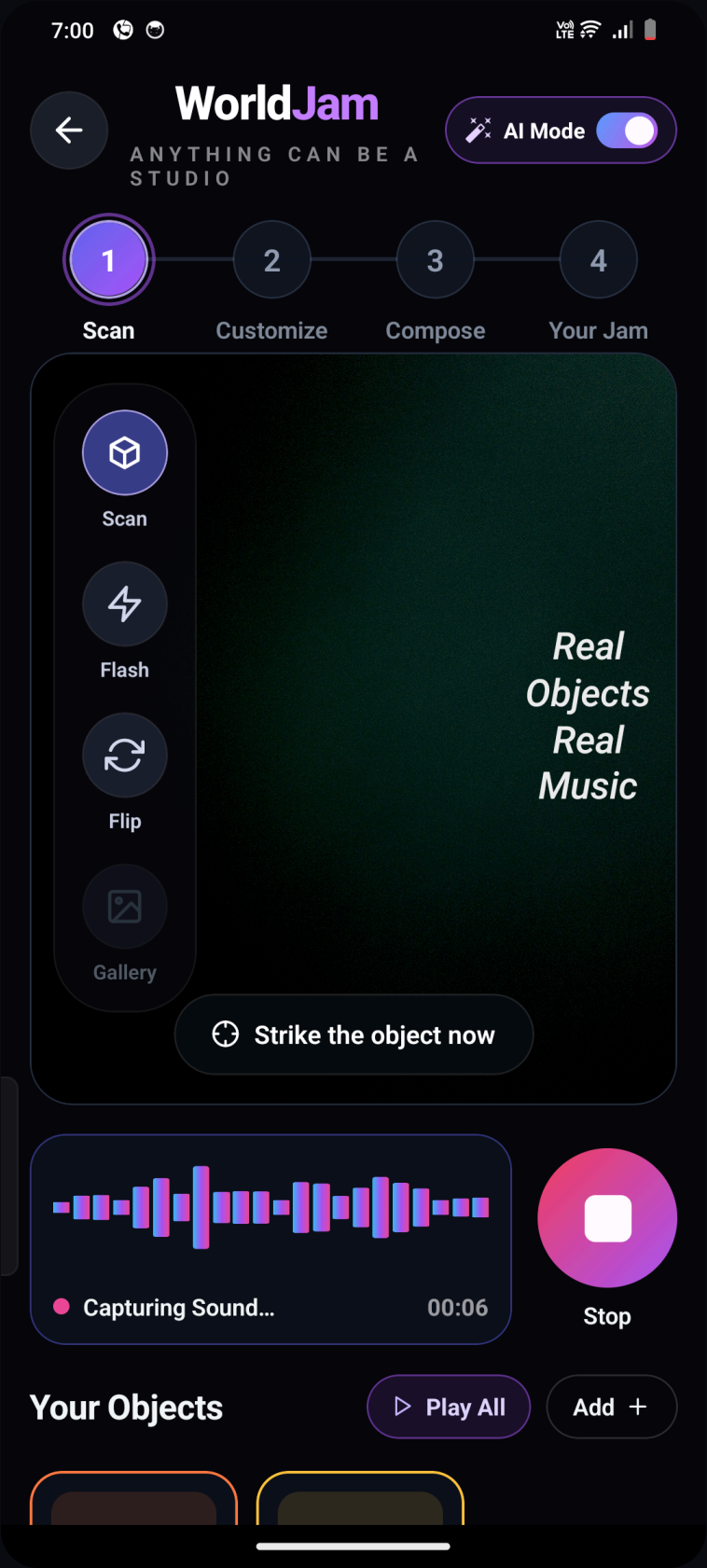
9

screens
shipped

0

bundled
sounds

The last number is the point: every sound in a WorldJam track is one the user recorded.



MEASURED ON DEVICE, NOT CLAIMED

24.8 ms from tap to sound

Below the ~50 ms threshold where a tap stops feeling like an instrument and starts feeling like a button. The whole product rests on this number, so the app measures it live — a judge can reproduce it on the spot.

Oboe exclusive-mode stream

Lock-free trigger ring buffer

Zero-allocation audio callback

Look-ahead scheduler

Three layers, each doing one job

Native audio

C++ and Oboe, 651 lines. A lock-free ring buffer carries taps to a callback that never allocates, locks or logs.

Signal analysis

FFT, YIN pitch tracking, onset detection and Krumhansl-Schmuckler key estimation decide what each recording should be.

On-device model

Gemma 4 E2B via llama.cpp emits a validated JSON plan. A rule-based arranger answers in microseconds if it is slow.

The fallback matters: the app is fully playable before the model has answered anything, and stays playable if no model is installed at all. The status line names which arranger produced the plan.

WHAT MAKES IT SOUND LIKE MUSIC

A loop repeats. A song goes somewhere.

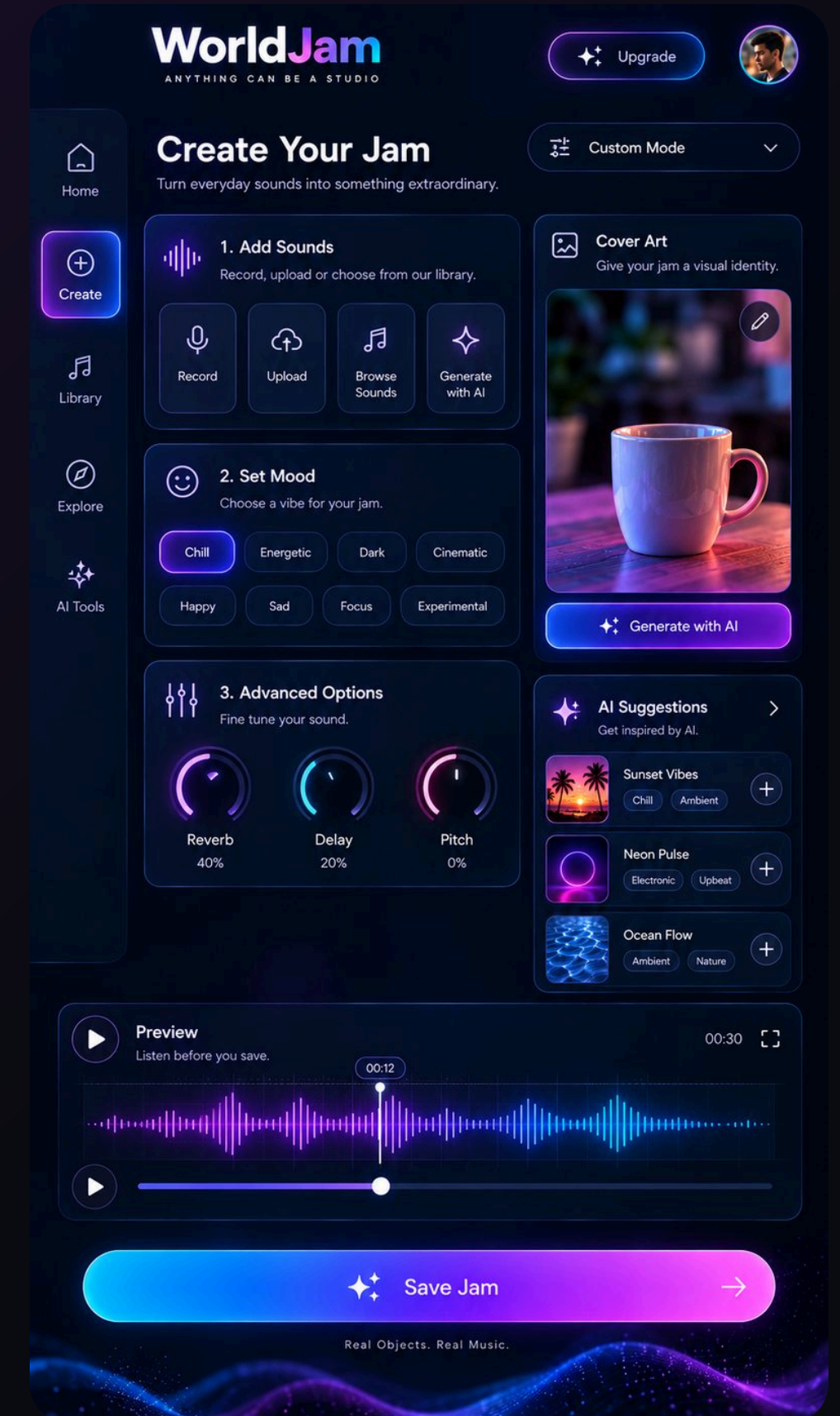
Sections with per-role density — A verse thins parts to their strong beats instead of muting them.

Velocity follows the bar — Flat velocity is the clearest sign something was sequenced, not played.

Fills on every turnaround — Quieter than the downbeat they lead into, so the arrival is not swallowed.

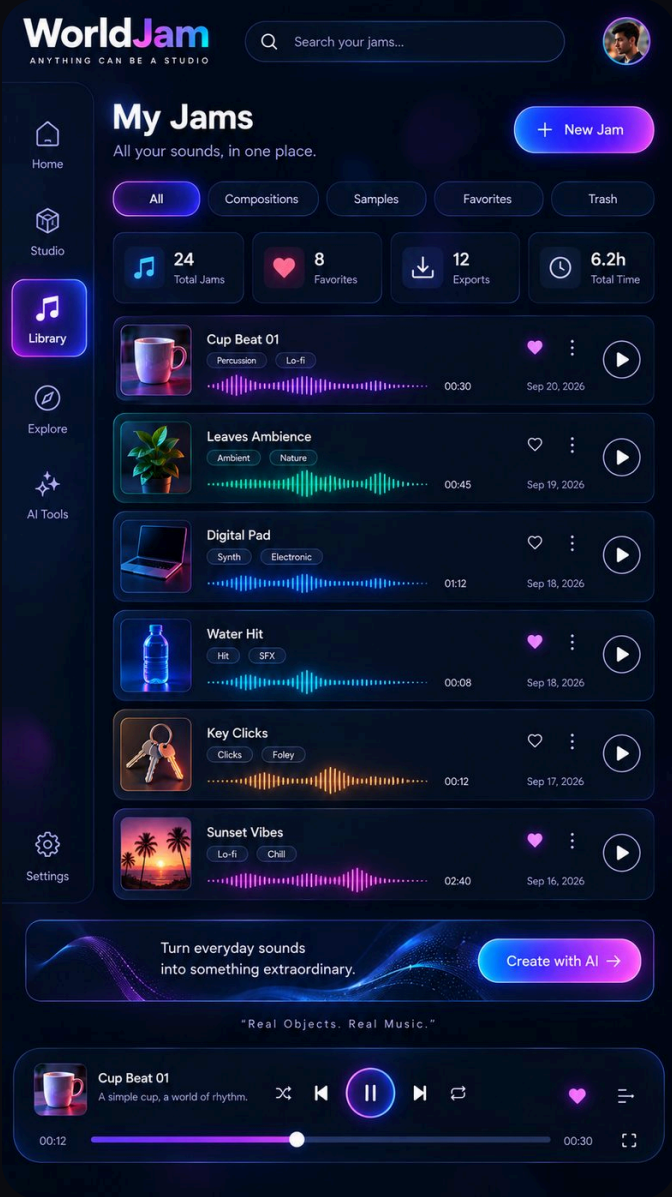
Your sung melody drives it — Hum a line: it is tuned, timed, and the arrangement is built around it.

No synthetic drums — Your objects are the percussion. A synth kit over them sounds canned.



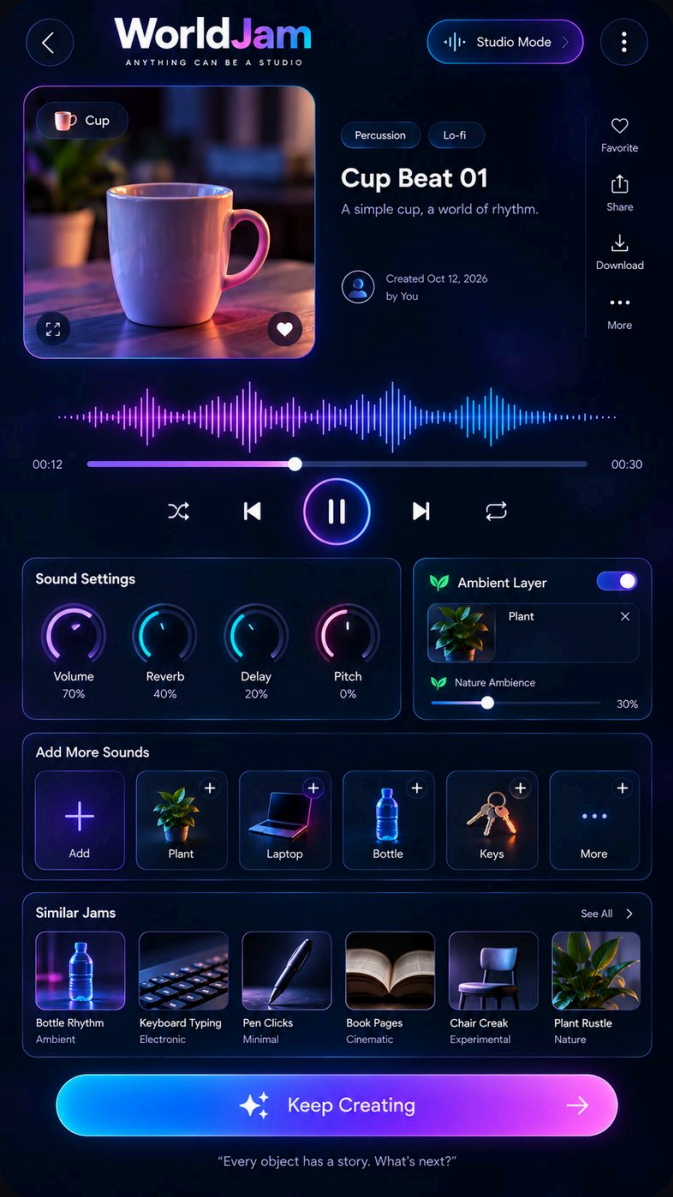
A finished product, not a demo path

Library, track detail, publishing and sharing are designed and built, so a jam has somewhere to live after it is made.



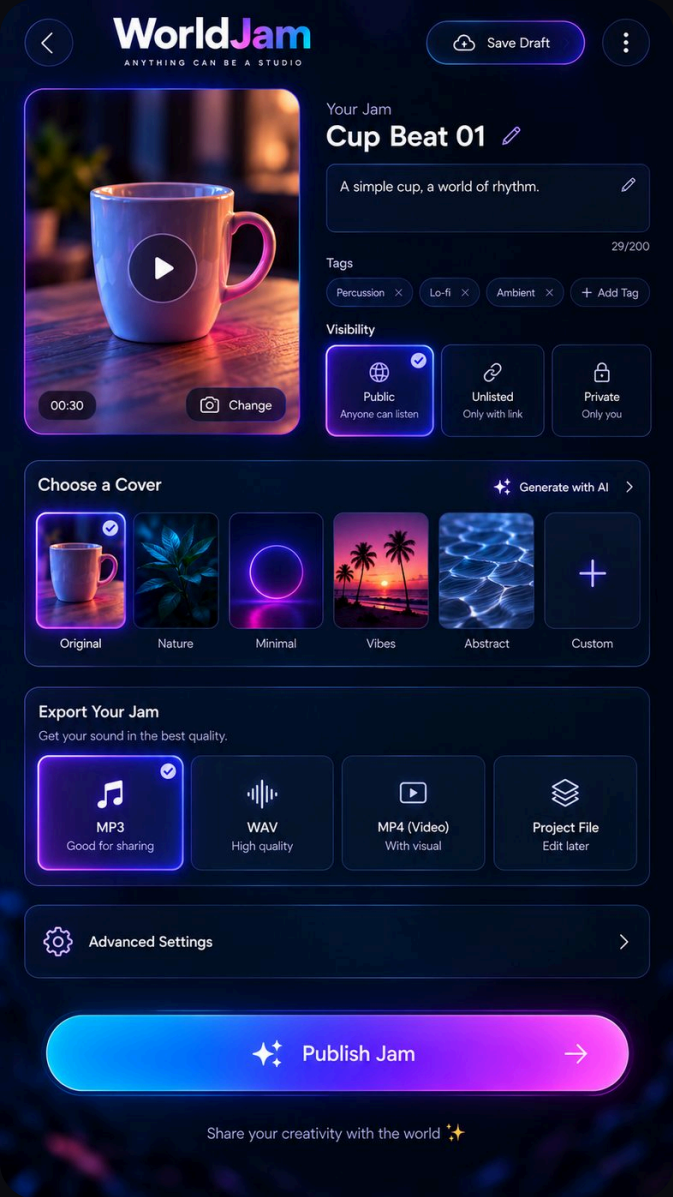
My Jams

Search, filter and stats over everything saved.



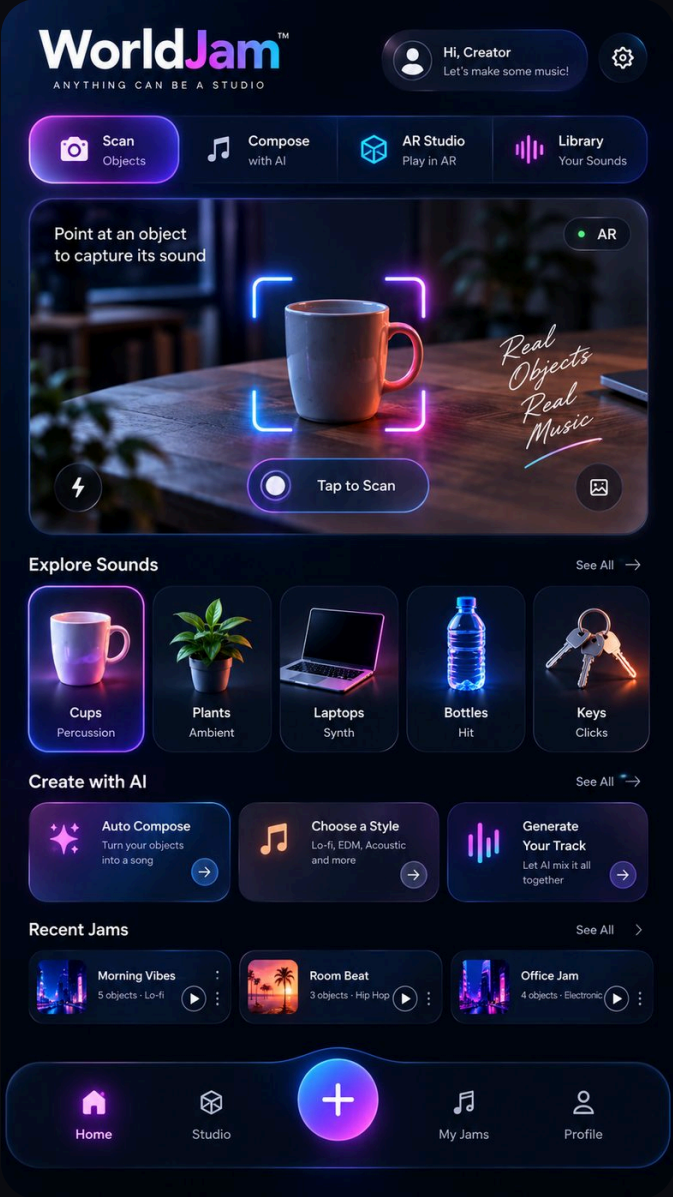
Track Detail

Transport, layers, ambience, similar jams.



Publish

Covers, visibility, MP3 / WAV / MP4 export.



Home

Explore sounds and pick up where you left off.

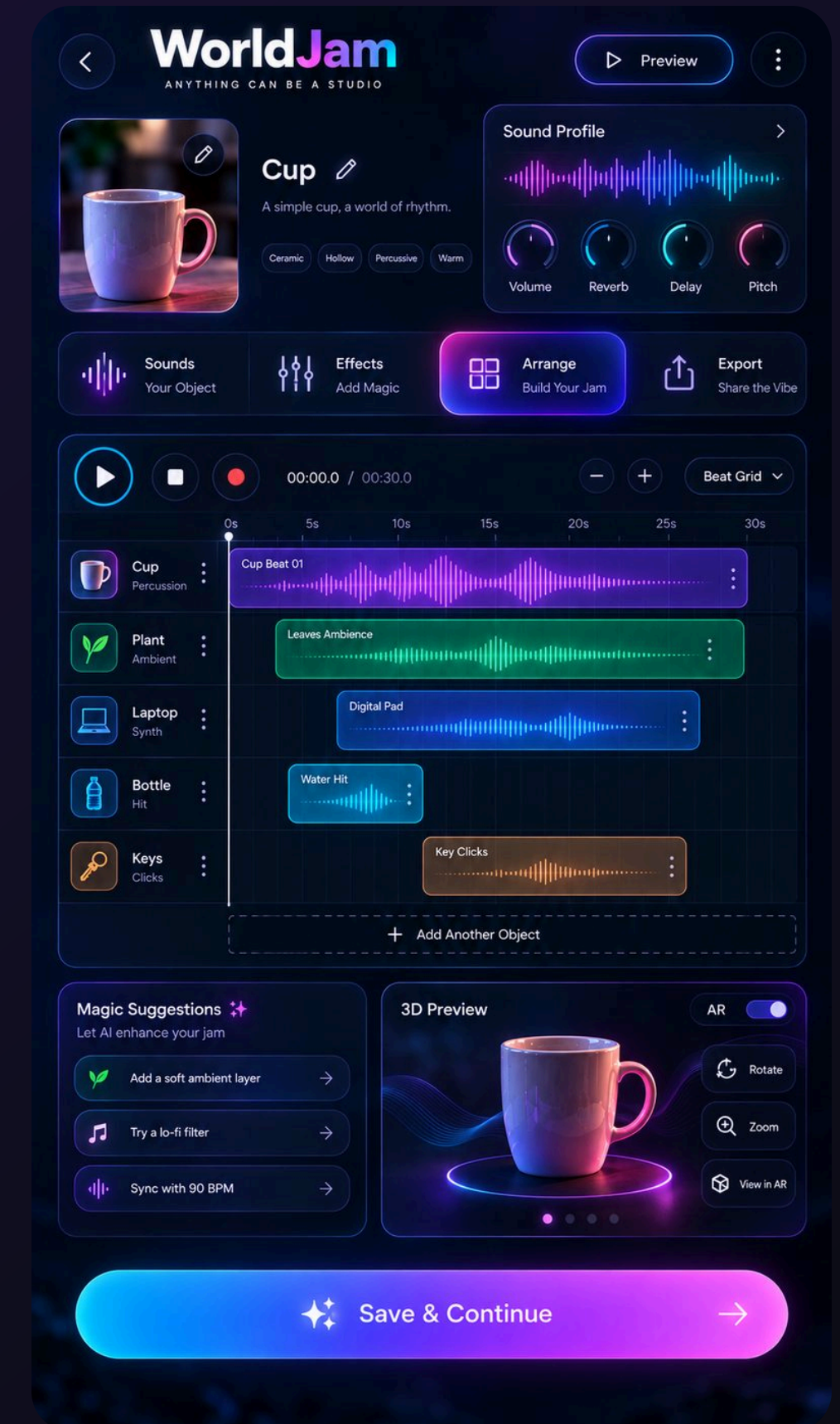
WHAT WE ARE BUILDING NEXT

AR Tiles — play the beat you just made

Your generated track becomes a game. Tiles rise through the room in AR, timed to the beats of *your* recording, and you tap them in the air with the phone. Every note you hit is the sound of your own cup, table or keys.

- The chart is generated from the arrangement, so every song plays differently.
- Scores reward timing against the 24.8 ms audio path already built.
- An audio-only mode: tiles announced by spatial sound, playable without sight.
- Share a jam and a friend plays your beat as a level.

The AR anchoring, spatial mixer and low-latency scheduler this needs are already in the build.



System architecture

Everything below runs on the handset. There is no server, and no code path that sends audio anywhere.

INPUT

Mic + Camera

AudioRecord at 48 kHz mono into a ring buffer; CameraX preview.

GATE

Transient window

Noise floor estimate, onset walk-back, tail gate. Keeps the hit and its ring.

ANALYSE

Feature extraction

FFT centroid, RMS decay, YIN pitch, tonality. Decides the musical role.

PLAN

Gemma 4 E2B

llama.cpp, 6 threads, 128-token JSON plan. Schema-validated and repaired.

RENDER

Arrangement

Form, densities, velocity curve, fills. One bar becomes a song.

OUT

Oboe callback

Lock-free trigger queue, 8-voice mixer, spatial pan. Never allocates.

Real-time thread is sacred

The audio callback never allocates, locks, logs or crosses the JS bridge. Triggers arrive through a lock-free SPSC queue; the scheduler pushes events 120 ms ahead so a GC pause cannot drop a beat.

The model is never in the loop

Inference happens on a button press, not per tap. If it is slow, missing or returns bad JSON, the rule-based arranger answers in microseconds and the UI says which one you heard.

React Native 0.76 (new architecture) · Expo SDK 52 dev build · custom Turbo-style native module · Zustand store · 17 Jest suites in CI-ready form.

WHERE THE WORK ACTUALLY WENT

Engineering decisions worth defending

C++ engine, not a library

Oboe in exclusive mode with the burst size the device prefers. Forcing 48 kHz inserts a resampler and drops you into shared mode — we measured both and kept the fast path.

Playback is the recording

Denoising runs only on the analysis copy. Spectral subtraction strips the quiet partials that make a cup sound like a cup, so the audio you hear is bit-identical to what was captured.

A contract, not a prompt

The model returns JSON validated against a schema: unknown objects dropped, beats clamped, BPM bounded. A malformed reply degrades to rules instead of reaching the audio engine.

Melody refinement

A hummed line is cleaned, snapped to scale by at most a semitone, timed from its own phrasing and octave-centred — gently, because over-correcting removes what made it human.

651

lines of C++ / JNI

20,256

lines of TypeScript

17

test suites

389

tests passing

0

network calls

Against the rubric

Criterion	Weight	What WorldJam brings
End product quality	30%	Runs on device now. 389 tests. Every screen reachable, every feature wired to real audio.
Novelty and impact	20%	No sample packs. The instrument is whatever is on your desk — and it works without sight or a network.
Creative phone use	15%	Microphone, camera, haptics and an on-device LLM are all load-bearing, not decoration.
Technical depth	15%	Custom C++ audio engine, lock-free scheduling, FFT and pitch analysis, a validated JSON contract with a fallback.
Office Kit usage	10%	Built and debugged phone-to-laptop throughout, with live device logs driving the fixes.
Demo and presentation	10%	The demo is the product: tap a cup on stage, then play the song it became.

The 25% read from device data is where an on-device model and real sensor use pay off — nothing there is self-reported.

WHAT HAPPENS ON STAGE

Hand us any object. We will make a song out of it.

Three minutes: record two objects from the judging table, hum eight bars, and play back the track the phone composed — with no connection and nothing pre-loaded. Then close your eyes and play it again.

Team PRXFR

WorldJam · Anything can be a studio